ECE 447

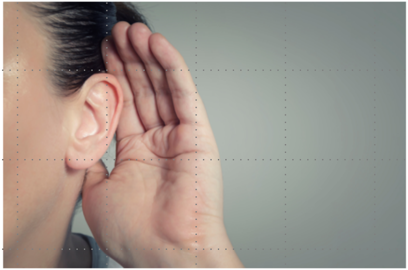
Lesson 06

Analysis and Transmission of Signals

Life/Leadership Lesson of the Day

- Listen!
- You aren't going to learn anything if you are the one doing all of the talking
- Plus, you run the risk of showing off what you don't know

- Good relationship advice, too!



Schedule and Admin

- Schedule
- Admin
  - **Lab 1 Assignment**. The assignment associated with Lab 1 is due Lesson 7 (Mon, 24 Aug) by 2359 via Gradescope upload.
  - **HW1**. Assigned today. Due Lesson 8 (Wed, 26 Aug).

Fourier Transform Pair

Synthesis (*Inverse Fourier Transform*):

$$x(t) = \mathcal{F}^{-1}[X(f)] = \int_{-\infty}^{\infty} X(f)e^{+j2\pi ft}df$$

Analysis (just *Fourier Transform*):

$$X(f) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt$$

When using angular frequency:

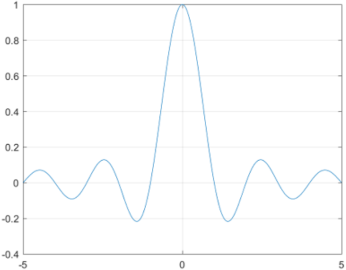
$$X(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

$$x(t) = \mathcal{F}^{-1}[X(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(f)e^{+j\omega t}d\omega$$

Fourier Transform Example

Rectangle Function, $\Pi(t/\tau)$

$$
\begin{aligned}
X(f) &= \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt \\
&= \int_{-\tau/2}^{+\tau/2} 1e^{-j2\pi ft}dt \\
&= \frac{-1}{j2\pi f}\left(e^{-j2\pi ft} - e^{j2\pi ft}\right)\Bigg|_{-\tau/2}^{+\tau/2} \\
&= \frac{\sin(\pi f\tau)}{(\pi f)} \\
&= \tau\text{sinc}(f\tau)
\end{aligned}
$$



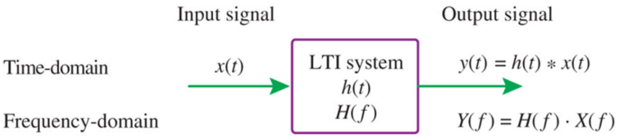
Bandwidth?

Fourier Transform Pairs and Properties

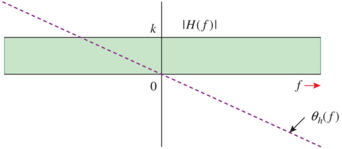
Tables 3.1 and 3.2 in textbook

- $\delta(t) \Leftrightarrow$
- $\delta(f - f_0) \Leftrightarrow$
- $\cos(2\pi f_0 t) \Leftrightarrow$
- $\Lambda(t) \Leftrightarrow$
- $C \Leftrightarrow$
- $g(at) \Leftrightarrow$
- $g(t)e^{j2\pi f_0 t} \Leftrightarrow$
- $g_1(t) * g_2(t) \Leftrightarrow$

LTI Systems

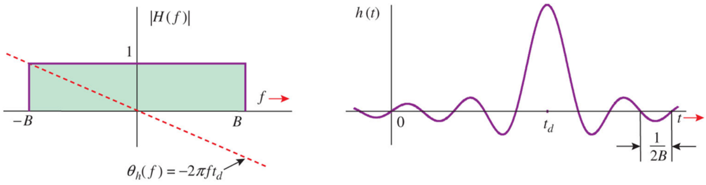


- $H(f) =$
- $|Y(f)| =$, $\theta_y(f) =$
- Distortionless transmission: $y(t) = k \cdot x(t - t_d)$
- $|H(f)| =$, $\theta_h(f) =$



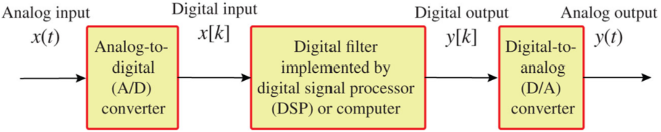
Filters

- Why can't we build ideal filters?



(a) (b)

- SDRs → Digital filters → DSP!



Channel Effects

- Linear distortion (amplitude or phase)
  - Spreads digital pulses out
  - Creates Inter Symbol Interference (ISI), or when a digital pulse smears outside of its time slot
- "Fading"
  - Large scale fading: changes in P_R based on stable conditions; caused by shadowing, blocking from buildings, topography
  - Small scale fading: rapid changes in P_R relative to signal freq; caused by multipath, mobility

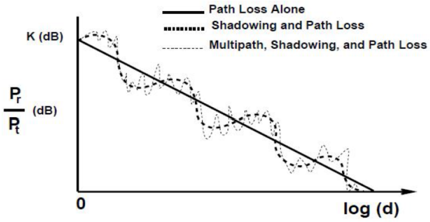


Fig.1.1 Path loss, shadowing and multipath versus distance

Parseval's Theorem and ESD

- Parseval's Theorem:

$$E_g = \int_{-\infty}^{\infty} |G(f)|^2 \, df$$

- Energy Spectral Density (ESD):

$$\Psi_g(f) = |G(f)|^2 \quad \rightarrow \quad E_g = \int_{-\infty}^{\infty} \Psi_g(f) df$$

- Use to determine essential bandwidth B (e.g., B that contains 95% of E_g - see Ex. 3.21... and HW problem 3.7-5)
- Modulation shifts ESD up and down, just like with a signal's spectrum; total energy is halved

Example: 95% Energy Bandwidth

Find the essential bandwidth B holding 95% of the energy in $g(t) = e^{-at}u(t), a > 0$.

- Spectrum, then ESD: $G(f) = \frac{1}{a+j2\pi f} \rightarrow \Psi_g(f) = |G(f)|^2 = \frac{1}{a^2+4\pi^2 f^2}$
- Total energy, easier in the time domain: $E_g = \int_0^\infty e^{-2at}dt = \frac{1}{2a}$
- Energy inside $|f| < B$:

$$E_B = \int_{-B}^{B} \frac{df}{a^2 + 4\pi^2 f^2} = \frac{1}{\pi a} \arctan\left(\frac{2\pi B}{a}\right)$$

- Set $E_B = 0.95\, E_g$: $\arctan\left(\frac{2\pi B}{a}\right) = 0.475\pi \rightarrow \frac{2\pi B}{a} = \tan(85.5^\circ) = 12.71$
- $B \approx 2.02a$ Hz. $G(f)$ never truly ends, but 95% of the energy is under $\approx 2a$ Hz

PSD and Autocorrelation

- Power Spectral Density (PSD):

$$S_g(f) = \lim_{T\to\infty} \frac{|G_T(f)|^2}{T} \quad \rightarrow \quad P_g = \int_{-\infty}^{\infty} S_g(f)df,$$

  where $G_T(f)$ is a truncated part of the power signal $G(f)$
- Time average of the ESD of $g_T(t)$
- Can also take Fourier Transform of the signal's *time autocorrelation* function $\mathcal{R}_g(\tau)$:

$$S_g(f) = \mathcal{F}[\mathcal{R}_g(\tau)]$$

- What is time autocorrelation? Next slide...

Time Autocorrelation

- Defined as:

$$\mathcal{R}_g(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t)g^*(t-\tau)dt$$

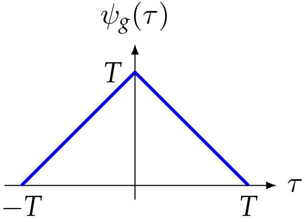
- This is an **average over time**, not a statistical average (that comes later in the course)
- Similar to convolution without the flip

Example: Autocorrelation of a Pulse

Let $g(t) = \Pi\left(\frac{t-T}{T}\right)$: unit height, width T, centered at $t = T$ (running from $\frac{T}{2}$ to $\frac{3T}{2}$).

- One pulse is an **energy** signal, so drop the $\frac{1}{T}$ time average: $\psi_g(\tau) = \int_{-\infty}^{\infty} g(t)g(t-\tau)dt$
- Slide the copy by τ: the pulses share a length $T - |\tau|$, and miss entirely once $|\tau| > T$

$$\psi_g(\tau) = \begin{cases} T - |\tau|, & |\tau| \leqslant T \\ 0, & \text{otherwise} \end{cases}$$



- A triangle of height T and base $2T$, i.e. $T\Delta\left(\frac{\tau}{2T}\right)$
- Two checks: $\psi_g(0) = T = E_g$, and we never used "centered at T" – autocorrelation **discards time shift**

Example: From Autocorrelation to the PSD

Now transform that triangle.

- Using $\Delta\left(\frac{\tau}{\tau_0}\right) \leftrightarrow \frac{\tau_0}{2}\text{sinc}^2\left(\frac{f\tau_0}{2}\right)$ with $\tau_0 = 2T$:

$$\Psi_g(f) = \mathcal{F}[\psi_g(\tau)] = T \cdot T\text{sinc}^2(fT) = T^2\text{sinc}^2(fT)$$

- Check it the direct way: $G(f) = T\text{sinc}(fT)e^{-j2\pi fT}$, so $|G(f)|^2 = T^2\text{sinc}^2(fT)$ – the $e^{-j2\pi fT}$ from centering at $t = T$ drops out, same lesson as before

- That is an **ESD**. One pulse has zero average power, so to get a **PSD** the pulse must recur – send one every T_0 seconds and average over the period:

$$S_g(f) = \frac{\Psi_g(f)}{T_0} = \frac{T^2}{T_0}\text{sinc}^2(fT)$$

- First null at $f = 1/T$: shorter pulses → wider spectrum